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Task-1:

To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
shashankreddy@Shashanks-MacBook-Air ~ % nano task1.c
```

```
#include <stdio.h>
#include <string.h>
int main()
{
  char command[100];
  while(1)
  {
    printf("Enter a command (type exit to quit): ");
    fgets(command, sizeof(command), stdin);
    command[strcspn(command, "\n")] = 0;
    if(strcmp(command, "exit") == 0)
    {
      printf("Exiting Program...\n");
      break;
    }
    printf("You entered: %s\n", command);
  }
  return 0;
```

Then press ctrl+oenter ctrl+x2)then gcc filename.c -o filename

```
shashankreddy@Shashanks-MacBook-Air ~ %  gcc task1.c -o task1
shashankreddy@Shashanks-MacBook-Air ~ %  ./task1
```

enter the linux command

```
shashankreddy@Shashanks-MacBook-Air ~ %  nano task1.c
```

```
shashankreddy@Shashanks-MacBook-Air ~ %  gcc task1.c -o task1
shashankreddy@Shashanks-MacBook-Air ~ %  ./task1
Enter a command (type exit to quit): hello
You entered: hello
Enter a command (type exit to quit): exit
Exiting Program...
shashankreddy@Shashanks-MacBook-Air ~ %
```

Design Control Flow Diagram

```
Start
  |
Display Prompt
  |
Read User Input
  |
Is Input "exit"?
  |      |
 No      Yes
  |      |
Display    Exit Program
Input      |
  |      |
  <---------
```

Task-2

Step 1: Create File1

```
nano File1.txt
```

Write:

Welcome to Ubuntu Lab

Step 2: Create File2

```
shashankreddy@Shashanks-MacBook-Air ~ %  nano file1.txt
shashankreddy@Shashanks-MacBook-Air ~ %  touch File2.txt
shashankreddy@Shashanks-MacBook-Air ~ %
```

Step 3: Create C Program

```
shashankreddy@Shashanks-MacBook-Air ~ %  nano file1.txt
shashankreddy@Shashanks-MacBook-Air ~ %  touch File2.txt
shashankreddy@Shashanks-MacBook-Air ~ %  nano copyfile.c
```

```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
  char buffer[100];
  int source, destination, bytes;
  // open() - Opens File1 for reading
  source = open("File1.txt", O_RDONLY);
  // open() - Opens File2 for writing
  destination = open("File2.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
  // read() and write() are used inside the loop
  while((bytes = read(source, buffer, sizeof(buffer))) > 0)
  {
    write(destination, buffer, bytes);
  }
  // close() - Closes both files
  close(source);
  close(destination);
  printf("File copied successfully.\n");
  return 0;
}
```

Compile and Run

```
shashankreddy@Shashanks-MacBook-Air ~ % nano file1.txt
shashankreddy@Shashanks-MacBook-Air ~ %  touch File2.txt
shashankreddy@Shashanks-MacBook-Air ~ %  nano copyfile.c
shashankreddy@Shashanks-MacBook-Air ~ %  gcc copyfile.c -o copyfile
shashankreddy@Shashanks-MacBook-Air ~ %  ./copyfile
```

OUTPUT

```
shashankreddy@Shashanks-MacBook-Air ~ %  nano file1.txt
shashankreddy@Shashanks-MacBook-Air ~ %  touch File2.txt
shashankreddy@Shashanks-MacBook-Air ~ %  nano copyfile.c
shashankreddy@Shashanks-MacBook-Air ~ %  gcc copyfile.c -o copyfile
shashankreddy@Shashanks-MacBook-Air ~ %  ./copyfile
File copied successfully.
shashankreddy@Shashanks-MacBook-Air ~ %
```